Unit 11, Lesson 5: Scatter Plots and Lines of Fit

Benchmarks

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

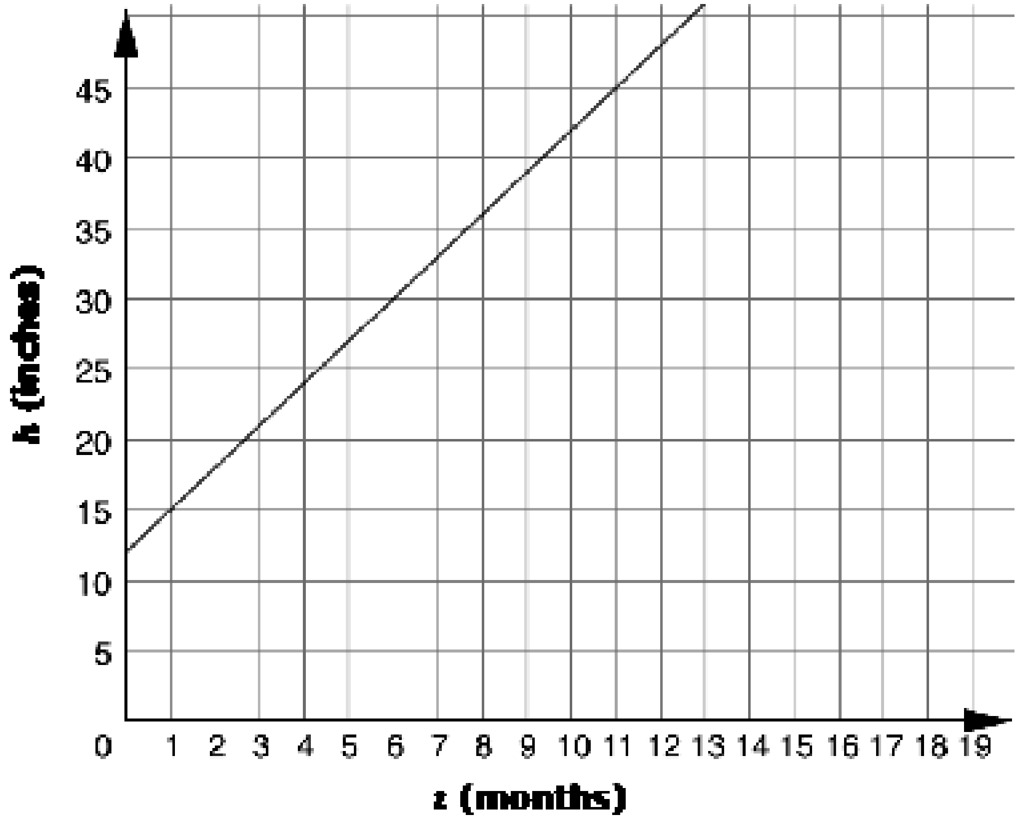
MA.8.DP.1.3 Given a scatter plot with a linear association, informally fit a straight line.

Learning Targets

- ☐ **I can describe patterns of associations in data on a scatter plot.**
- ☐ **I can informally fit a straight line to data on a scatter plot.**

11.5.1 Warm-Up: Bell Work

1. The graph shows the height in inches, *h*, of a bamboo plant *t* months after it has been planted.



a. What is the slope of the line?

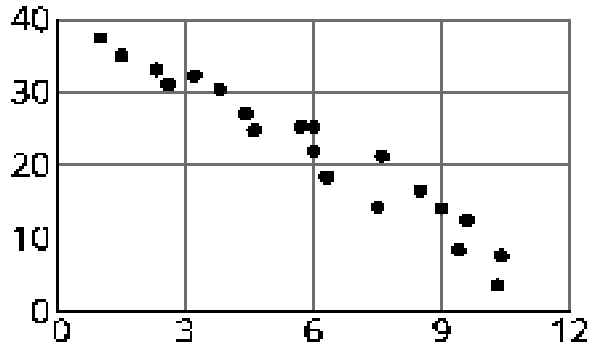
b. Interpret the slope in this context.

c. The *y*-intercept of the line is 12. Interpret the meaning of the *y*-intercept in context.

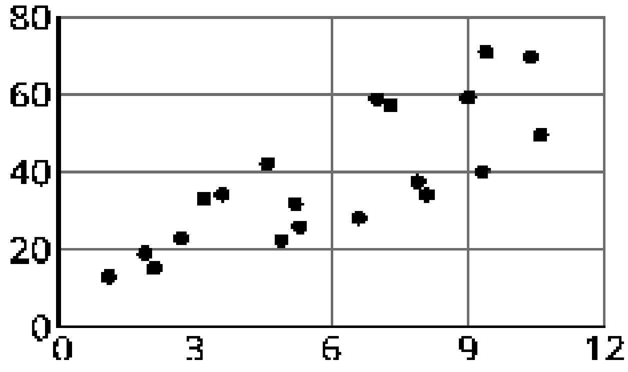
11.5.2 Collaboration: Patterns of Association

1. Four scatter plots are shown.

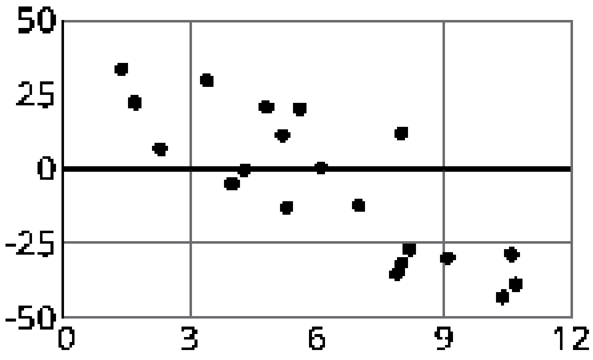
Scatter Plot A



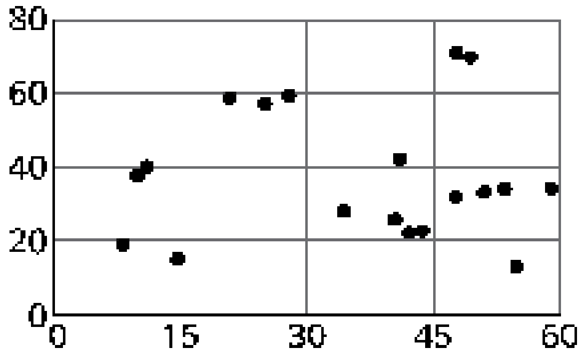
Scatter Plot B



Scatter Plot C



Scatter Plot D



a. Describe the direction of the linear association between the two variables in each scatter plot.

Scatter Plot	Direction of the Association
A	
B	
C	
D	

b. Explain which of the scatter plots does not belong based on the strength of the linear association between the two variables.

c. Discuss with your partner whether any of the scatter plots appear to have any outliers.

d. Use a straightedge to draw a line that models the linear relationship between the two variables on each scatter plot.

e. Compare your lines with your partner's. Discuss any differences and make revisions to your lines, if necessary.

f. Explain which of the scatter plots does not belong based on the y-intercept of the lines you drew to model the linear relationships.

g. Explain which of the scatter plots does not belong based on the slope of the lines you drew to model the linear relationships.

11.5.3 Activity: Informally Fitting Lines to Data

1. Your teacher will give you and your partner a scatter plot and information about the relationship it models.

   a. Record the letter of the provided scatter plot.

   Scatter Plot ________

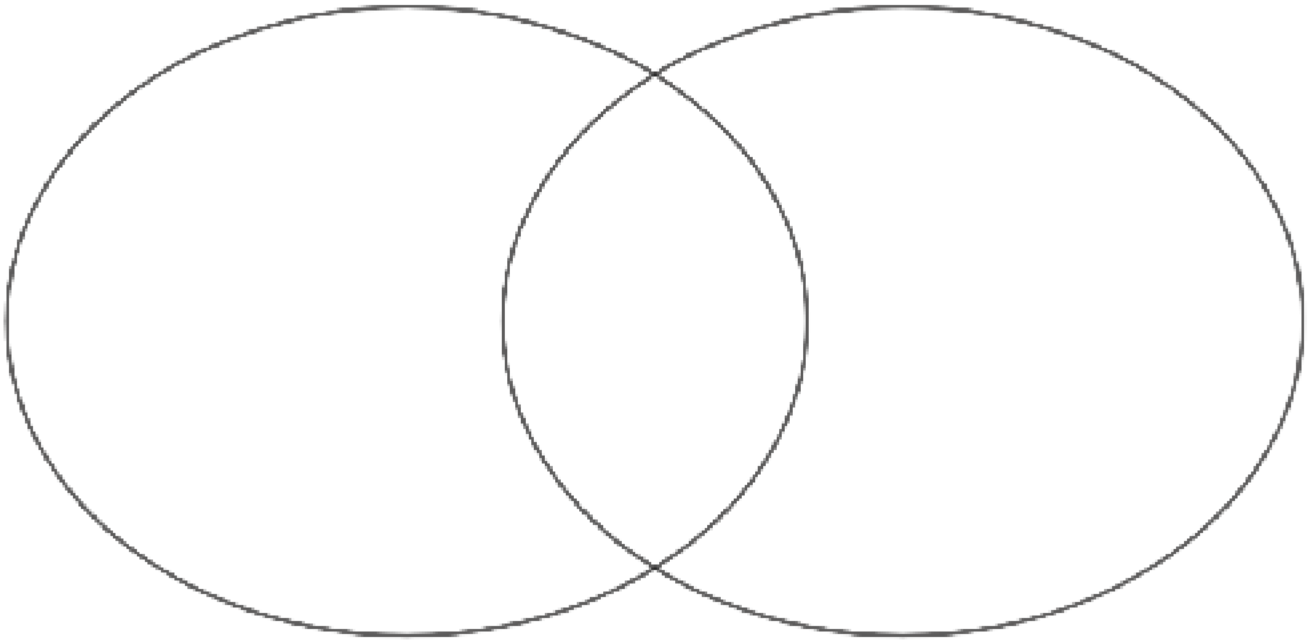
   b. Describe any patterns of association.

   c. Work with your partner to use a straightedge to draw a line that models the relationship between the two variables on the scatter plot.

   d. Explain how the slope of your line of fit relates to the patterns of association described in part b.

2. Find a group with the same lettered scatter plot as you.

   a. Compare and contrast both groups' work. Record your observations on the Venn diagram.

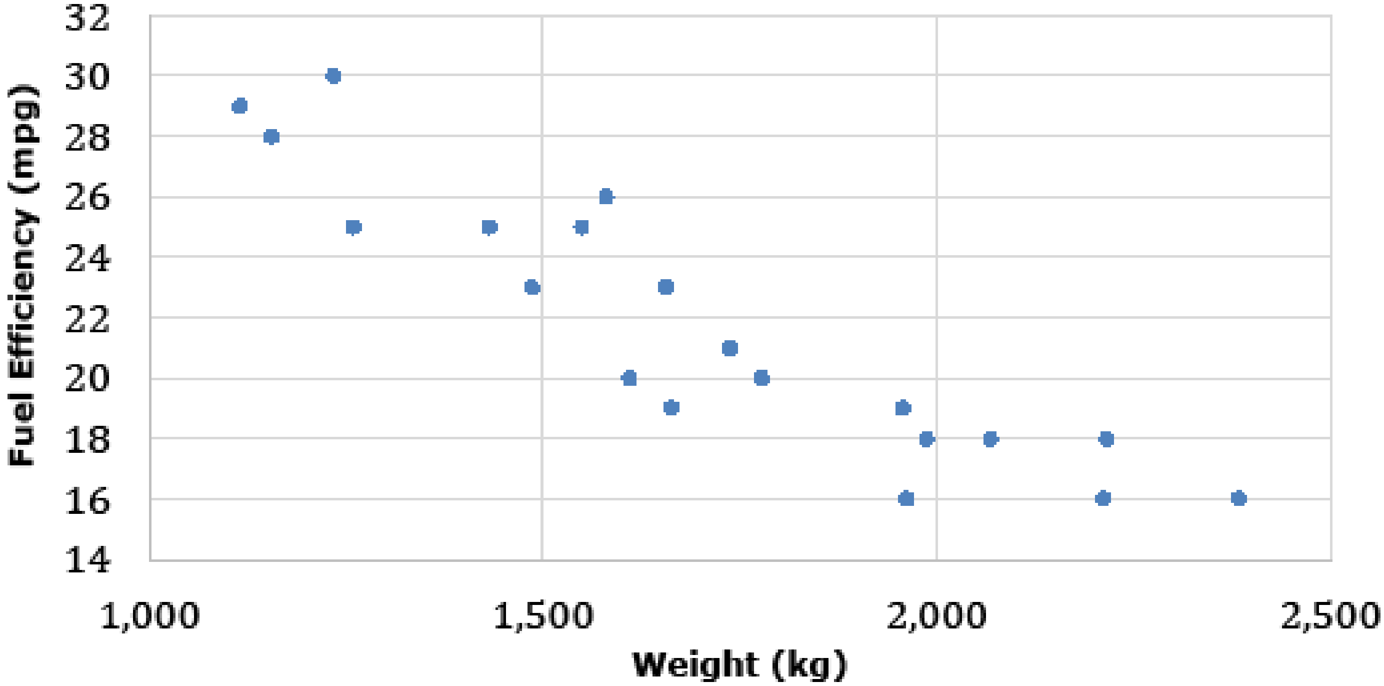


   b. Both groups had the same scatter plot. How would you explain any differences in your work?

   c. Explain which group's line fits the data better.

11.5.4 Wrap-Up: Ticket Out the Door

1. The scatter plot shows the relationship between a car's weight in kilograms (kg) and its fuel efficiency in miles per gallon (mpg).



a. Informally fit a line to the data using a straightedge.

b. Describe how the slope of your line relates to the strength and direction of the association between the weight and fuel efficiency of a car.